

Executive Summary: PODZO provides a centralized operational interface for managing SAP-connected contracts and purchase orders, transport allocation, delivery execution, proof-of-delivery (POD) verification, and freight invoicing through to SAP MIRO supplier invoice parking and payment clearing.

SECTION 1 — PODZO AT A GLANCE

PODZO is an enterprise operational execution platform designed to bridge SAP S/21 with day-to-day transport logistics and bulk haulage fulfillment. Used by industrial shippers, commercial hauliers, siding supervisors, drivers, and receiving customers, PODZO manages the physical realization of transport purchase orders. By retrieving active contracts and released POs directly from SAP S/21, PODZO governs vehicle allocation, 4-point weighbridge validation, digital journey tracking, and OCR-assisted POD verification—delivering end-to-end operational visibility while preserving SAP as the single authoritative financial core.

✓ Direct SAP S/21 Alignment

Live ingestion of active contracts & transport POs via real-time OData / REST services.

✓ 4-Point Weight Governance

Tare & gross weighing at siding & receiving yard eliminates payload disputes.

✓ Automated Settlement & MIRO

OCR verification feeds verified tonnage directly into SAP MIRO invoice parking.

SECTION 2 — KEY CAPABILITIES

◆ SAP S/21 Synchronization

Direct ingestion of active outline agreements (ME33K) and transport purchase orders (ME23N) with materials, target tonnage, agreed rates, and validity windows.

◆ Transport Allocation & E-Sign

Timebound PO allocation to certified transporters with mandatory digital signature acknowledgment and electronic carrier waybill (bilty) assignment.

◆ 4-Point Weighbridge Control

Captures dispatch tare/gross (loading siding) and destination tare/gross (receiving yard) to guarantee net payload accuracy and detect transit weight loss.

◆ Driver Mobile Execution

Mobile-first driver web console for route navigation, arrival timestamp logging, gate check-in, and instant camera capture of stamped physical POD slips.

◆ OCR POD Verification Desk

Automated document data extraction matching waybill numbers, truck registrations, and delivered weights against SAP records with confidence scoring and exception review.

◆ Invoicing & SAP MIRO Posting

Consolidation of verified delivered tonnage into freight invoices, staging audit-ready data for SAP MIRO supplier invoice parking, posting, and AP settlement.

SECTION 3 — ROLE-BASED OPERATIONS

CA

Company Admin

Manages contracts, PO quotas, variance review queues, and final MIRO invoice approvals.

TA

Transporter Admin

Accepts allocated POs via digital signature, schedules fleet/drivers, and submits freight invoices.

SR

Supervisor

Conducts siding inspections, logs weighbridge weights, and authorizes gate dispatch release.

DR

Driver

Executes assignments via mobile web, verifies loading, and uploads stamped physical POD slips.

CR

Customer

Captures receiving weighbridge readings, records transit deviations, and confirms delivery acceptance.

SECTION 4 — END-TO-END OPERATIONAL WORKFLOW

STEP 1

SAP S/21 Sync
Contracts & POs



STEP 2

PO E-Acceptance
Transporter (TA)



STEP 3

Siding Dispatch
Supervisor (SR)



STEP 4

Transit Execution
Driver (DR)



STEP 5

Yard Receiving
Customer (CR)



STEP 6

OCR POD Review
Company Admin (CA)



STEP 7

SAP MIRO Clear
Park & Post Invoice

Operational Data Continuity: SAP S/21 Contracts & POs • Transporter Digital E-Acceptance & Bilty • Siding Weighbridge Dispatch Clearance • Driver Mobile Telemetry • Yard Weighbridge Receiving • OCR 3-Way Match Verification • SAP MIRO Supplier Invoicing & Payment Clearing.

SECTION 5 — SAP S/21 INTEGRATION

Integration Dimension	Data Scope & Functional Mapping	Data Governance & Storage Boundary
Inbound from SAP S/21 OData / REST APIs	Outline Contracts (ME33K), Transport POs (ME23N), Line Items, Target Tonnage, Unit Rates, Cost Centers, Delivery Sites, Vendor Master (LFA1).	Retrieved from SAP. Master records remain governed exclusively in SAP S/21; cached for operational display.
Operational Processing PODZO Core Layer	Digital PO acceptance signature hashes, carrier waybills (bilties), 4-point weighbridge tickets, OTP verifications, driver stamps, uploaded POD scans.	Stored in PODZO. Field execution logs and audit telemetry are persisted locally in PODZO's transactional database.
Outbound to SAP S/21 ABAP RFC / MIRO API	Verified delivered quantities, Service Entry Sheet (SES / ML81N) triggers, MIRO supplier invoice parking/posting, ArchiveLink POD attachments.	Posted to SAP. Final financial and compliance artifacts are transmitted back to SAP for settlement.

SECTION 6 — HIGH-LEVEL SOLUTION ARCHITECTURE

1. Enterprise ERP Layer (SAP S/21 System of Record)

SAP ECC / S/4HANA

Outline Agreements (ME33K)
 Purchase Orders (ME23N)
 Service Entry Sheets (ML81N)
 Supplier Invoices (MIRO)
 ArchiveLink / DMS

↕ Secure SAP Integration Layer (REST • OData • RFC Proxy • Sync Logs) ↕

2. PODZO Operational Core Platform (Execution & Verification Layer)

Express • Node.js • SQLite WAL

4-Point Weighbridge Engine
 OCR Document Matching Desk
 Tolerance & Variance Engine
 Exception Review Queue
 Secure Document Vault

Audit & Transit Event Logger

↕ Role-Governed Secure Web Endpoints (JWT Authentication • RBAC) ↕

3. Role-Tailored Operational Consoles (SAP Fiori-Inspired UX)

React 19 • Desktop • Mobile

CA: Admin Control Tower & Approvals
 TA: Transporter POs & Fleet Operations
 SR: Siding Weighbridge Console
 DR: Driver Journey Web App

CR: Customer Receiving Yard

SECTION 7 — TECHNOLOGY OVERVIEW

FRONTEND React 19 • Vite TypeScript, React Router 7	BACKEND SERVICES Node.js • Express 5 RESTful API Architecture	SAP INTEGRATION S21 OData / REST OData, RFC Proxy, Sync Logs	DATA STORE SQLite (WAL Mode) Full Transactional Durability	DESIGN SYSTEM SAP Fiori Inspired Responsive Desktop & Mobile
--	--	---	---	---

SECTION 8 — BUSINESS VALUE

Unified Operational Truth
 Bridges SAP S/21 with field transporters and receiving yards without requiring third-party hauliers to hold full enterprise SAP user licenses.

Dispute-Free Weight Verification
 Mandatory 4-point weighbridge logging captures tare and gross weights at both origin and destination, eliminating payload discrepancies and transit loss disputes.

Compressed Invoicing Cycles
 Immediate digital POD capture paired with OCR verification replaces multi-week physical paper transit, collapsing billing and MIRO clearing to hours.

Complete Audit Governance
 Every vehicle dispatch, weighbridge entry, tolerance override, and MIRO posting is immutably timestamped with user credentials and audit logs.

SOLUTION APPROACH & POSITIONING
 PODZO provides a purpose-built operational execution and verification layer around SAP-connected contract and purchase order workflows. It enables business users, logistics providers, and drivers to collaborate through a modern, role-based interface while preserving strict alignment, security, and traceability with SAP S/21 business data.